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$NaSO_2Ph$ Induced Exclusive Isomerization of 1,2-Allenyl Sulfoxides to 2-Alkynylic Phenyl Sulfoxides

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The clean isomerization of 1,2-allenyl sulfoxides in the presence of $NaSO_2Ph$ afforded 2-alkynylic phenyl sulfoxides in good yields and selectivities. Although strong base-catalyzed isomerization of allenes to alkynes is known, the mechanism for the current reaction is interesting for the obvious reason that $NaSO_2Ph$ is a weak base.

Keywords 1,2-allenyl sulfoxide, isomerization, 2-alkynic phenyl sulfoxide.

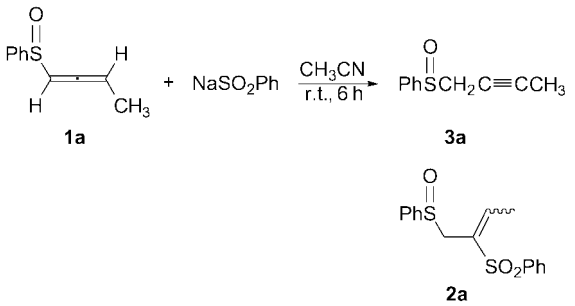
Allenes are a class of compounds with unique reactivity.[1] Although these compounds had been considered highly unstable for a long period of time,[2] recently much attention has been paid to the chemistry of allenes. This has even become a hot current research area in organic chemistry.[3]

Heteroatom-substituted allenes have shown excellent regio- and stereo-selectivity in halohydroxylation reactions.[4,5] We observed that the stereochemistry of halohydroxylation of allenyl sulfides is different from that of allenyl sulfoxides. Actually 1,2-allenyl sulfoxides are a class of allenes, which can undergo both electrophilic addition and nucleophilic addition. Due to the weak electron-withdrawing ability of the sulfoxides, $AlCl_3$ was used as a mediator to promote the hydrohalogenation of 1,2-allenyl sulfoxides.[6] Continuing our study on the nucleophilic addition reaction of 1,2-allenyl sulfoxides, we wish to report here the $NaSO_2Ph$-mediated clean isomerization of 1,2-allenyl sulfoxides to propargyl sulfoxides.

Initially we tested the reaction of 1,2-butadienyl sulfoxide **1a** with $NaSO_2Ph$ in MeCN in order to prepare **2a**. To our surprise instead of the nucleophilic conjugate addition product **2a**, we observed that 2-butynyl phenyl sulfoxide **3a** was formed as the only product. With 2 equiv. of $NaSO_2Ph$ **3a** was isolated in 82% yield (Scheme 1). The reaction could proceed with 10 mol% of $NaSO_2Ph$ affording **3a** in 74% yield.

Although there have been many reports on the clean isomerization from alkynes to allenes,[7-9] those on the isomerization of allenes to alkynes are quite limited.[10] Thus, we studied this reaction in detail. The results of solvent effect shown in Table 1 indicate that acetonitrile is the best solvent.

Scheme 1



$NaSO_2Ph$ (equiv.)	0.1	0.2	0.5	1.0	2.0
Yield of **3a**/%	74	74	75	76	82

Table 2 lists some typical examples of this isomerization. It should be noted that (1) the reaction is general for 3-monosubstituted 1,2-allylic sulfoxides. (2) The yields range from moderate to good. (3) The conversion is complete since no starting **1** was recovered after the time specified in Table 2.

Strong base-catalyzed isomerization reaction of allenes to alkynes has been well documented in literature,[10] with the mechanism shown in Scheme 2. It is quite interesting to note that $NaSO_2Ph$ is a weak base, which may not be able to deprotonate the starting material **1** in the current transformation. Thus, the mechanism is a still a puzzle to us.

In conclusion, a clean conversion of allenes to alkynes has been observed. Compared to potassium 3-aminopropylamide or *N*-methylbutylamide,[10] the

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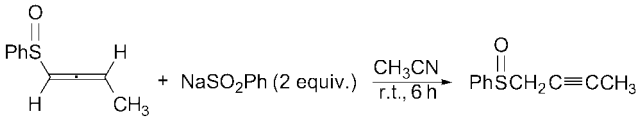
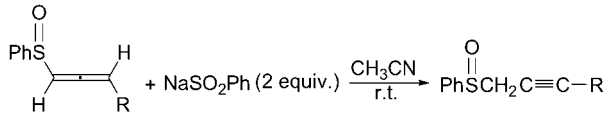
Table 1 Reaction of $NaSO_2Ph$ with 1,2-allenyl phenyl sulfoxides in different solvents



1a **3a**

Entry	Solvent	Isolated yield of **3a**/%	**1a** recovered/%
1	Me_2CO	59	0
2	THF	60	0
3	DMF	60	0
4	DMSO	67	0
5	PhH	8	57
6	PhMe	24	43
7	MeCN	83	0

Table 2 Reaction of $NaSO_2Ph$ with 1,2-allenyl phenyl sulfoxides[a]

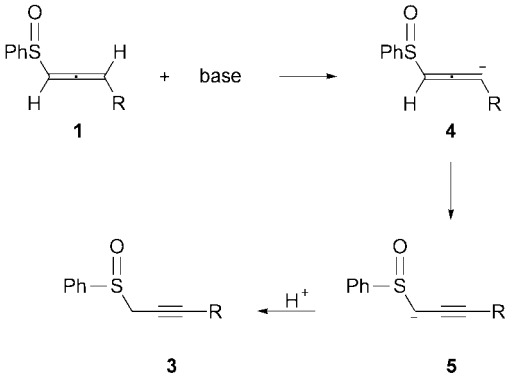


1 **3**

Entry	R	Time/h	Yield of **3**[b]/%
1	CH_3 (**1a**)	6	83 (**3a**)
2	C_2H_5 (**1b**)	6	85 (**3b**)
3	*i*-C_3H_7 (**1c**)	21	84 (**3c**)
4	*n*-C_4H_9 (**1d**)	22	86 (**3d**)
5	*n*-C_5H_{11} (**1e**)	5	87 (**3e**)
6	*n*-C_7H_{15} (**1f**)	21	92 (**3f**)
7	*n*-C_9H_{19} (**1g**)	6	77 (**3g**)
8	Ph (**1h**)	19	55 (**3h**)[c]

[a] The reaction was carried out using 0.3—0.5 mmol of 1,2-allenyl phenyl sulfoxides. [b] Isolated yield. [c] 9% of **1h** was recovered.

Scheme 2



lower basicity and easy availability of $NaSO_2Ph$ make the current transformation synthetically and mechanistically interesting. Further study in this area is being conducted in our laboratory.

Experimental

The starting materials **1** were prepared according to the known procedure.[11]

Typical procedure

A solution of **1a** (71.6 mg, 0.4 mmol) and $NaSO_2Ph$ (131.2 mg, 0.8 mmol) in 2 mL of MeCN was stirred at 16 ℃ for 6 h. The mixture was quenched with 5 mL of water, extracted with diethyl ether (25 mL×3), washed with brine, and dried over anhydrous Na_2SO_4. Evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate=3∶1, *V*∶*V*) afforded 59.3 mg (83%) of 2-butynyl phenyl sulfoxide (**3a**).[11] Oil. ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.69—7.71 (m, 2H), 7.53—7.54 (m, 3H), 3.65 (dq, *J*=16, 2 Hz, 1H), 3.56 (dq, *J*=16, 2 Hz, 1H), 1.80 (t, *J*=2 Hz, 3H); ^{13}C NMR ($CDCl_3$, 100 MHz) δ: 143.1, 131.4, 128.8, 124.3, 84.4, 67.7, 48.4, 3.5; IR (KBr) *v*: 2236, 1582, 1477, 1443, 1404, 1050 cm^{-1}; MS (70 eV, EI) *m/z* (%): 178 (M^+, 53.94), 43 (100).

Synthesis of 2-pentynyl phenyl sulfoxide (3b)

The reaction of 76.8 mg (0.4 mmol) of **1b** and 137.7 mg (0.8 mmol) of $NaSO_2Ph$ afforded 65.0 mg (85%) of **3b**: oil. ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.71—7.73 (m, 2H), 7.53—7.55 (m, 3H), 3.71 (d, *J*=16 Hz, 1H), 3.56 (d, *J*=16 Hz, 1H), 2.15—2.20 (m, 2H), 1.08 (t, *J*=7.6 Hz, 3H); ^{13}C NMR ($CDCl_3$, 100 MHz) δ: 143.1, 131.3, 128.7, 124.4, 90.3, 67.9, 48.4, 13.4, 12.3; IR (KBr) *v*: 2237, 1586, 1443, 1050 cm^{-1}; MS (70 eV, EI) *m/z* (%): 192 (M^+, 22.35), 135(100). HRMS calcd for $C_{11}H_{12}OSNa^+$ 215.0501, found 215.0503.

Synthesis of 4-methyl-2-pentynyl phenyl sulfoxide (3c)

The reaction of 80.6 mg (0.4 mmol) of **1c** and 131.9 mg (0.8 mmol) of $NaSO_2Ph$ afforded 67.6 mg (84%) of **3c**: oil. ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.70—7.72 (m, 2H), 7.53—7.54 (m, 3H), 3.71 (d, *J*=16 Hz, 1H), 3.55 (d, *J*=16 Hz, 1H), 2.47—2.53 (m, 1H), 1.02—1.05 (m, 6H); ^{13}C NMR ($CDCl_3$, 100 MHz) δ: 143.0, 131.4, 128.7, 124.5, 94.4, 67.7, 48.3, 22.5, 20.4; IR (KBr) *v*: 2227, 1583, 1511, 1466, 1048 cm^{-1}; MS (70 eV, EI) *m/z* (%): 206 (M^+, 28.84), 135 (100). HRMS calcd for $C_{12}H_{14}OS$ 206.07654, found 206.07329.

Synthesis of 2-heptynyl phenyl sulfoxide (3d)[12]

The reaction of 88.6 mg (0.4 mmol) of **1d** and 132.6 mg (0.8 mmol) of $NaSO_2Ph$ afforded 75.8 mg (86%) of **3d**: oil. ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.70—7.72 (m, 2H), 7.52—7.53 (m, 3H), 3.71 (d, *J*=16 Hz, 1H), 3.59 (d, *J*=16 Hz, 1H), 2.14—2.17 (m, 2H), 1.38—1.45 (m, 2H), 1.27—1.35 (m, 2H), 0.88 (t, *J*=6.8 Hz, 3H); ^{13}C NMR ($CDCl_3$, 100 MHz) δ: 143.2, 131.3, 128.7, 124.4, 89.0, 68.4, 48.4, 30.2, 21.6, 18.3, 13.4; IR (KBr) *v*: 2298, 1670, 1583, 1465, 1444, 1052 cm^{-1}; MS (70 eV, EI) *m/z* (%): 220 (M^+, 30.77), 135 (100).

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Synthesis of 2-octynyl phenyl sulfoxide (3e)[12]

The reaction of 95.0 mg (0.4 mmol) of **1e** and 132.1 mg (0.8 mmol) of $NaSO_2Ph$ afforded 81.7 mg (87%) of **3e**: oil. ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.64—7.66 (m, 2H), 7.45—7.48 (m, 3H), 3.63 (dt, J=16, 2 Hz, 1H), 3.53 (dt, J=16, 2 Hz, 1H), 2.07—2.11 (m, 2H), 1.35—1.39 (m, 2H), 1.19—1.23 (m, 4H), 0.82 (t, J=4 Hz, 3H); ^{13}C NMR ($CDCl_3$, 100 MHz) δ: 143.1, 131.3, 128.7, 124.4, 89.0, 68.4, 48.4, 30.7, 27.9, 21.9, 18.6, 13.7; IR (KBr) v: 2235, 1583, 1444, 1087, 1052 cm^{-1}; MS (70 eV, EI) m/z (%): 235 (M^++1, 57.39), 135 (100).

Synthesis of 2-decynyl phenyl sulfoxide (3f)[12]

The reaction of 80.4 mg (0.3 mmol) of **1f** and 100.3 mg (0.6 mmol) of $NaSO_2Ph$ afforded 73.6 mg (92%) of **3f**: oil. ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.71—7.73 (m, 2H), 7.53—7.55 (m, 3H), 3.71 (d, J=16 Hz, 1H), 3.60 (d, J=16 Hz, 1H), 2.14—2.17 (m, 2H), 1.38—1.42 (m, 2H), 1.18—1.36 (m, 8H), 0.90 (t, J=7.2 Hz, 3H); ^{13}C NMR ($CDCl_3$, 100 MHz) δ: 143.2, 131.3, 128.8, 124.4, 89.1, 68.4, 48.5, 31.6, 28.6, 28.3, 22.5, 18.7, 13.9; IR (KBr) v: 2231, 1583, 1444, 1088, 1053 cm^{-1}; MS (70 eV, EI) m/z (%): 263 (M^++1, 62.4), 41 (100).

Synthesis of 2-dodynyl phenyl sulfoxide (3g)

The reaction of 72.6 mg (0.25 mmol) of **1g** and 84.1 mg (0.5 mmol) of $NaSO_2Ph$ afforded 55.8 mg (77%) of **3g**: oil. ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.70—7.72 (m, 2H), 7.53—7.54 (m, 3H), 3.70 (dt, J=16, 2 Hz, 1H), 3.58 (dt, J=16, 2 Hz, 1H), 2.13—2.16 (m, 2H), 1.39—1.44 (m, 2H), 1.26—1.31 (m, 12H), 0.89 (t, J=6 Hz, 3H); ^{13}C NMR ($CDCl_3$, 100 MHz) δ: 143.3, 131.4, 128.7, 124.5, 89.2, 68.5, 48.6, 31.8, 29.4, 29.2, 29.1, 28.7, 28.4, 22.6, 18.8, 14.1; IR (KBr) v: 2231, 1583, 1444, 1088, 1053 cm^{-1}; MS (70 eV, EI) m/z (%): 290 (M^+, 49.68), 41 (100). HRMS calcd for $C_{18}H_{26}OS$ 290.17044, found 290.16961.

Synthesis of 3-phenyl-2-propynyl phenyl sulfoxide (3h)[12]

The reaction of 96.4 mg (0.4 mmol) of **1h** and 133.4 mg (0.8 mmol) of $NaSO_2Ph$ afforded 8.8 mg of **1h** and 53.2 mg (55%) of **3h**: oil. ^{1}H NMR ($CDCl_3$, 400 MHz) δ: 7.77—7.79 (m, 2H), 7.55—7.57 (m, 3H), 7.29—7.34 (m, 5H), 3.96 (d, J=16 Hz, 1H), 3.84 (d, J=16 Hz, 1H); ^{13}C NMR ($CDCl_3$, 100 MHz) δ: 143.1, 131.8, 131.7, 129.0, 128.8, 128.3, 124.6, 122.1, 88.2, 78.1, 49.0; IR (KBr) v: 2242, 1598, 1490, 1444, 1087, 1052 cm^{-1}; MS (70 V, EI) m/z (%): 240 (M^+, 52.14), 115 (100).

References

1 (a) Schuster, H. F.; Coppola, G. M. *Allenes in Organic Synthesis*, John Wiley & Sons, New York, **1984**.
(b) Patai, S. In *the Chemistry of Ketenes, Allenes, and Related Compounds*, Part 1, John Wiley & Sons, New York, **1980**.
(c) Lipshutz, B. H. *Chem. Rev*. **1986**, *86*, 795.

2 (a) Burton, B. S.; Pechmann, H. V. *Chem. Ber*. **1887**. *20*, 145.
(b) Jones, E. R. H.; Mansfield, G. H.; Whiting, M. L. J. *J. Chem. Soc*. **1954**, 3208.

3 (a) Zimmer, R.; Dinesh, C. U.; Nadanan, E.; Hhan, F. A. *Chem. Rev.* **2000**, *100*, 3067.
(b) Reissig, H. U.; Schade, W.; Amombo, G. M. O.; Pulz, R.; Hausherr, A. *Pure Appl. Chem.* **2002**, *74*, 175.
(c) Ma, S.; Li, L. *Synlett.* **2001**, 1206.
(d) Ma, S. *Acc. Chem. Res*. **2003**, *36*, 701.
(e) Ma, S. M. In *Handbook of Organopalladium Chemistry for Organic Synthesis*, Eds.: Negishi, E.; De Meijer, A., Wiley-Interscience, New York, **2002**, p. 1491.

4 (a) Ma, S.; Wei, Q.; Wang, H. *Org. Lett.* **2000**, *2*, 3893.
(b) Ma, S.; Ren, H.; Wei, Q. *J. Am. Chem. Soc.* **2003**, *125*, 4817.

5 (a) Ma, S.; Hao, X.; Huang, X. *Org. Lett.* **2003**, *5*, 1217.
(b) Ma, S.; Hao, X.; Huang, X. *Chem. Commun*. **2003**, 1082.
(c) Fu, C.; Ma, S.; Huang, X. *Tetrahedron. Lett.* **2004**, *45*, 6063.

6 Ma, S.; Wei, Q. *Eur. J. Org. Chem*. **2000**, 1939.

7 For typical examples reporting the isomerization from functionalized alkynes to allenes, see:
(a) Majetich, G.; Zhong, Y.; Dreyer, G. *Tetrahedron Lett*. **1993**, *34*, 449.
(b) Jacobs, T. L.; Brill, W. F. *J. Am. Chem. Soc.* **1953**, *75*, 1314.
(c) Eglinton, G.; Jones, E. R. H.; Mansfield, G. H.; Whiting, M. C. *J. Chem. Soc.* **1954**, 3197.
(d) Truce, W. E.; Markley, L. D. *J. Org. Chem.* **1970**, *35*, 3275.
(e) Reiche, H. J.; Shah, S. K.; Gold, P. M.; Olson, R. E. *J. Am. Chem. Soc.* **1981**, *103*, 3112.
(f) Sturtz, G.; Paugam, J. P.; Corbil, B. *Synthesis* **1974**, 730.
(g) Kanematsu, K.; Kinoyama, I. *J. Chem. Soc., Chem. Commun.* **1992**, 735.
(h) Yeo, S. K.; Shiro, M.; Kanematsu, K. *J. Org. Chem.* **1994**, *59*, 1621.
(i) Padwa, A.; Caruso, T.; Nahm, S.; Rodriguez, A. *J. Am. Chem. Soc.* **1982**, *104*, 2865.
(j) Boggini, G.; Brucé, L.; Zecchi, G. *J. Chem. Soc., Perkin Trans. 1* **1990**, 533.
(k) Chen, Y. S. P.; Dominguez, E.; Garratt, P. J.; Neoh, S. B. *Tetrahedron Lett.* **1978**, 691.
(l) Pourcelot, G.; Cadiot, P. *Tetrahedron* **1982**, *38*, 2123.
(m) Brandsma, L.; Wejers, H. E.; Arens, J. F. *Recueil* **1963**, *82*, 1040.
(n) Wei, L.-L.; Xiong, H.; Douglas, C. J.; Hsung, R. P. *Tetrahedron Lett*. **1999**, *40*, 6903.

8 For isomerization of propargylic ethers to 1,2-allenyl ether, see:
(a) Castro, J.; Moyano, A.; Pericàs, M. A.; Riera, A.; Greene, A. E.; Alvarez-Larena, A.; Piniella, J. F. *J. Org. Chem.* **1996**, *61*, 9016 and references cited therein.
(b) Kadota, I.; Kawada, M.; Gevorgyan, V.; Yamamoto, Y. *J. Org. Chem.* **1997**, *62*, 7439.
(c) Dulcere, J. P.; Mihoui, M. N.; Rodriuez, J. *J. Chem. Soc., Chem. Commun.* **1988**, 237.
(d) Braverman, S.; Duar, Y.; Segv, D. *Tetrahedron Lett.*

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1976, 3181.
(e) Fall, Y.; Gomez, G.; Fernandez, C. *Tetrahedron Lett.* **1999**, *40*, 8307.
(f) Tius, T. A.; Ousset, J. B.; Astrab, D. P.; Fauq, A. H.; Trehan, S. *Tetrahedron Lett.* **1989**, *30*, 923.
(g) Tius, M. A.; Hu, H. *Tetrahedron Lett.* **1998**, *39*, 5937.
(h) Hayakawa, K.; Ohsuki, S.; Kanematsu, K. *Tetrahedron Lett.* **1986**, *27*, 947.
(i) Moghaddm, F. M.; Emami, R. *Synth. Commun*. **1997**, *27*, 4073.
(j) Dulcère, J. P.; Crandall, J.; Faure, R.; Santelli, M.; Agati, V.; Mihoubi, M. N. *J. Org. Chem.* **1993**, *58*, 5702.
(k) Dulcère, J. P.; Mihoubi, J. M. N.; Rodiguez, J. *J. Org. Chem.* **1993**, *58*, 5709.
(l) Rochet, P.; Vatèle, J. M.; Goré, J. *Synthesis* **1994**, 795.

9 For isomerization between simple alkynes and allenes, see:
(a) Jacobs, T. L.; Awawie, R.; Cooper, R. G. *J. Am. Chem. Soc.* **1951**, *73*, 1273.
(b) Hopf, H. *Chem. Ber*. **1971**, *104*, 3087.
(c) Gensler, W. J.; Casella, J. *J. Am. Chem. Soc.* **1958**, *80*, 1376.
(d) Huntsman, W. D.; Boer, J. A.; Woosley, M. H. *J. Am. Chem. Soc.* **1966**, *88*, 5846.
(e) Cordes, J. F.; Günzler, H. *Chem. Ber.* **1959**, *92*, 1053.

10 For reports on the isomerization from allenes to alkynes, see:
(a) Spence, J. D.; Wyatt, J. K.; Bender, D. M.; Moss, D. K.; Nantz, M. H. *J. Org. Chem.* **1996**, *61*, 4014.
(b) Grattan, T. J.; Whitehurst, J. S. *J. Chem. Soc., Perkin Trans. 1* **1990**, 11.

11 Baudin, J. B.; Julia, S. A.; Lorne, R. *Bull. Soc. Chim. Fr*. **1992**, 440.

12 Fortes, C. C.; Garrote, C. F. D. *Synth. Commun*. **1997**, *27*, 2993.

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